

Abstract

The global healthcare landscape is currently facing a significant challenge in managing musculoskeletal (MSK) disorders, which have become a leading driver of healthcare costs and physical disability. Traditional physical therapy models often fail due to barriers such as high costs, geographical limitations, and low patient adherence to home exercise programs (HEPs). This project introduces the **Range of Motion (ROM) Tracker**, a next-generation virtual physical therapy platform designed to bridge the gap between clinical supervision and home rehabilitation.

The primary objective of this study is to develop a "Zero-Hardware" solution that utilizes advanced Computer Vision (CV) to perform markerless motion capture. Built using the **Unity** development engine and **Google MediaPipe Pose**, the system identifies and tracks 33 skeletal landmarks in real-time using only a standard smartphone camera. This approach eliminates the need for expensive proprietary sensors while maintaining clinical-grade accuracy for monitoring therapeutic progress.

The core innovation of the ROM Tracker lies in its "**Provider Enablement**" model. Unlike existing digital clinics that replace traditional therapists, this platform empowers independent clinics by providing them with the tools to offer Remote Therapeutic Monitoring (RTM). The system features a specialized **RTM Billing Engine** that automatically tracks data transmission and provider review time, facilitating reimbursement through specific CPT codes. Furthermore, the project addresses health equity by implementing a **Senior-Centric UX**, incorporating voice control, high-contrast interfaces, and tablet optimization to cater to the elderly demographic most affected by MSK issues.

To ensure patient privacy, the system utilizes **Edge AI** architecture, where video processing occurs locally on the device and is immediately discarded, transmitting only anonymous skeletal data to the cloud. This "Privacy-by-Design" approach ensures compliance with strict healthcare regulations such as HIPAA and GDPR.

Keywords:

Computer Vision; Physical Therapy; Range of Motion; Remote Therapeutic Monitoring; Edge AI; Unity.

Acknowledgement

First and foremost, I would like to express my deepest gratitude to my graduation project supervisor, **Dr. Mohammed Elmogy**, for their invaluable guidance, constant encouragement, and insightful feedback throughout the development of this project. Their expertise in the field of Information Technology was instrumental in shaping the technical architecture and clinical relevance of the ROM Tracker.

I would also like to thank the faculty members of the **Information Technology Department** at the **Faculty of Computers and Information, Mansoura University**, for providing a stimulating academic environment and the necessary resources to carry out this research.

Special thanks go to the physical therapists and healthcare professionals who participated in the requirements gathering phase, providing the clinical insights essential for building a tool that truly serves the needs of both patients and providers.

Finally, I am profoundly grateful to my family and teammates for their unwavering support and patience during this challenging journey. Their belief in this project's vision was the primary motivation behind its completion

List of Abbreviations

- AI: Artificial Intelligence
- AR: Augmented Reality
- B2B: Business-to-Business
- CPT: Current Procedural Terminology
- CV: Computer Vision
- DMSK: Digital Musculoskeletal
- ERD: Entity Relationship Diagram
- GDPR: General Data Protection Regulation
- HEP: Home Exercise Program
- HIPAA: Health Insurance Portability and Accountability Act
- HPE: Human Pose Estimation
- IMU: Inertial Measurement Unit
- JSON: JavaScript Object Notation
- MAE: Mean Absolute Error
- MSK: Musculoskeletal
- ROM: Range of Motion
- RTM: Remote Therapeutic Monitoring
- SaaS: Software-as-a-Service
- SDK: Software Development Kit
- TFLite: TensorFlow Lite
- UI: User Interface
- UML: Unified Modeling Language

- URP: Universal Render Pipeline
- UX: User Experience

Chapter 1: Introduction

1.1 Overview/Background

The global healthcare system is currently navigating a profound crisis in the management of musculoskeletal (MSK) disorders, which have become a primary driver of healthcare spending. The traditional "brick-and-mortar" clinic model for physical therapy suffers from insurmountable barriers to access, including geographical limitations and provider shortages. Consequently, adherence to prescribed home exercise programs (HEPs) is notoriously low, leading to chronic pain and unnecessary surgical interventions. This project proposes a "Range of Motion (ROM) Tracker" that leverages consumer-grade mobile devices and advanced computer vision to democratize access to clinical-grade biomechanical analysis.

1.2 Motivations and Problem Statement

- **High Costs of Existing Solutions:** Current digital musculoskeletal (DMSK) solutions often rely on expensive wearable sensors (IMUs) that are difficult to ship and maintain.
- **Adherence Gap:** Patients often struggle to perform exercises correctly at home without real-time guidance, risking re-injury.
- **Provider Struggle:** Independent physical therapy clinics are losing revenue to large digital disruptors and need tools to offer digital monitoring to their own patients.

1.3 Project Objectives

- To develop a mobile application that uses a standard smartphone camera to measure the range of motion of specific joints.
- To provide objective, data-driven insights that allow therapists to track patient recovery over time.
- To enable clinics to generate new revenue streams through Remote Therapeutic Monitoring (RTM) billing codes.

1.4 Contributions of This Study

- **Zero-Hardware Deployment:** Eliminates the need for proprietary sensors by using pure software efficiency.

- **Privacy by Design:** Utilizes Edge AI (On-Device Processing) to ensure patient video never leaves the device.
- **Provider Enablement:** Positions the tool as a partner for existing clinics rather than a replacement.

1.5 Document Organization

- **Chapter 1** introduces the project's motivations, objectives, and scope.
- **Chapter 2** provides a literature review of existing MSK digital solutions.
- **Chapter 3** details the system analysis, including requirements and architecture.
- **Chapter 4** describes the system design, including UI/UX and core algorithms.

1.6 Summary

This chapter provided a comprehensive overview of the current challenges in musculoskeletal (MSK) care, highlighting the economic burden and accessibility barriers of traditional therapy. It introduced the ROM Tracker as a strategic solution that leverages Edge AI and computer vision to provide a zero-hardware, privacy-first alternative for remote rehabilitation. By defining the project's objectives and contributions, this chapter established the foundation for a platform that empowers independent providers and improves patient adherence.

Chapter 2: Literature Review

2.1 Introduction

A project must be put in the context of existing works relevant to the topic to define how the investigation differs from other studies.

2.2 Review of Relevant Work

- **Sensor-Based Systems (e.g., SWORD Health, Hinge Health):** These pioneers use wearable inertial measurement units (IMUs) to capture data. While accurate, they have high friction due to hardware shipping and setup complexity.
- **Computer Vision Platforms (e.g., Kaia Health, Exer Health):** These use "markerless motion capture" via standard cameras. Kaia Health focuses on B2C/Employer markets, while Exer Health focuses on provider enablement.

2.3 Relationship Between Relevant Work and Our Work

Our ROM Tracker follows the "Provider Enablement" model of Exer Health but adds a specific focus on senior-centric UX design. While competitors target large employers, we empower local, independent clinics to retain their patients.

2.4 Summary

This chapter reviewed the evolution of digital musculoskeletal care, categorizing existing works into sensor-based and computer vision systems. While sensor-based pioneers like Hinge Health offer high accuracy, they face scalability issues due to hardware logistics. The ROM Tracker distinguishes itself by adopting a "Provider Enablement" model, focusing on senior-centric design and supporting local clinics rather than competing with them, thus filling a critical gap in the current market.

Chapter 3: System Analysis

3.1 Analysis of Existing Systems

The traditional model relies on paper handouts or simple videos, providing no mechanism for monitoring adherence or correcting form. Existing digital clinics (Hinge/SWORD) act as competitors to local therapists.

3.2 System Requirements

- **Functional Requirements:** Real-time skeletal tracking, joint angle calculation, session data logging, and a therapist dashboard.
- **Non-Functional Requirements:** Low latency (Edge AI), high data privacy (GDPR/HIPAA compliance), and accessibility for older adults.

3.3 System Architecture

The system uses a modular "Edge AI" architecture where video is processed directly on the phone's RAM and discarded immediately. Only anonymous skeletal coordinates are sent to the cloud.

3.4 Development Methodology

We utilize Unity (2D URP) as the development engine due to its superior mobile and AR support. Google's MediaPipe Pose is used for human pose estimation, providing a rich 33-landmark skeletal model.

3.5 Summary

This chapter detailed the analytical phase of the project, identifying the functional and non-functional requirements essential for a clinical-grade monitoring tool. It established the technical framework using Unity for mobile and AR support and MediaPipe Pose for detailed skeletal tracking. Furthermore, the modular Edge AI architecture was outlined to ensure that video processing remains on-device, prioritizing patient privacy and low-latency feedback.

Chapter 4: System Design

4.1 Introduction

The design explains how the study was formulated to calculate joint angles and visualize progress.

4.2 Data base Design or Algorithms

The core logic utilizes Vector Mathematics in C#. Specifically, the `Vector3.Angle()` function determines the angle between three body points (e.g., hip, knee, and ankle) to measure flexion.

4.3 Interface Design

The UX is designed with "Senior-Centric" principles:

- **Tablet-First Mode:** Optimized for larger screens preferred by older users.
- **Voice Control:** Allows interaction (e.g., "Start Exercise") without touching the screen while in position.
- **Simple Mode:** High-contrast buttons and step-by-step text instructions.

4.4 Summary

This chapter described the design specifications of the ROM Tracker, focusing on the mathematical models and user experience. The core logic was defined through vector mathematics to accurately calculate joint flexion, while the interface design prioritized the needs of the elderly through voice control and high-contrast, tablet-optimized layouts. These design choices ensure that the platform is not only clinically valid but also accessible and engaging for its target demographic.